

Lead AI Engineer · LLM Systems & Production GenAI · GCP · Azure · GPU Infra

Lead AI engineer and data scientist with 10+ years of engineering experience across classical ML, deep learning, forecasting, operations research, and LLM platforms. Built config-driven LoRA fine-tuning and self-hosted vLLM serving at Best Buy, replacing paid APIs and saving \$300K/year. Delivered retail fraud inference at 40,000+ events/sec under a 200 ms SLA. Leads the design and cross-functional delivery of AI/ML platforms, mentors engineers, and supports technical hiring.

Skills

- **AI/ML & Analytical Methods:** Data Science; Classical ML; Logistic Regression; Deep Learning; Machine Learning; LLM Fine-tuning; LoRA / QLoRA; LLM Evaluation; Autoencoders; Behavioral Anomaly Detection; Computer Vision; Transfer Learning; Time-Series Forecasting; Feature Engineering; Vehicle Routing; Mixed-Integer Programming; Operations Research; Constraint Modelling; Supply-Chain Simulation
 - **Frameworks & Libraries:** vLLM Serving; Unsloth; Hugging Face Transformers / PEFT; NVIDIA Morpheus; XGBoost; PyTorch; TensorFlow; Keras; TensorFlow Serving; Google OR-Tools; Pandas; NumPy; RAPIDS cuDF / CuPy
 - **GPU Computing:** Multi-GPU Training; GPU-Accelerated Computing; Distributed Model Training
 - **Cloud & MLOps:** Self-hosted LLMs; LLMOps; Google Cloud Platform; Microsoft Azure; GKE; Dataflow / Apache Beam; Vertex AI; Stream Processing; Pub/Sub; GCS; Artifact Registry; Azure ML Studio; Azure DevOps; Kubernetes; Helm; Istio; Docker; Seldon Core; Cloud Build; ArgoCD; Google KCC; Grafana; Prometheus; MLflow; Model Monitoring; CI/CD; GitOps
 - **Languages & Data:** Python; SQL; MS SQL Server; Power BI; Linux & Shell Scripting; Git
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Experience

Lead Artificial Intelligence Engineer

Best Buy | Jul 2025 - Present

- **LLM Fine-tuning Framework:** Built a config-driven LoRA framework using Unsloth, PEFT, and QLoRA, achieving ~2x faster training and 60-80% lower VRAM use. Supported single- and multi-GPU runs, multi-task adapters, and automated evaluation. YAML configuration enabled reusable tasks without custom Python; standardized adapter artifacts enabled shared vLLM serving without a separate deployment per task.
- **LLM Serving & Impact:** Deployed OpenAI-compatible vLLM endpoints on GKE with HPA and Istio; integrated GCS, Artifact Registry, Cloud Build, and Prometheus/Grafana. Self-hosting eliminated paid API dependency, saved \$300K/year, and strengthened data governance and compliance posture.
- **Retail Fraud Detection & Prevention:** Built Apache Beam/Dataflow streaming and GKE inference for XGBoost fraud models at 40,000+ events/sec under a 200 ms SLA. Covered payment, refund, and reship signals; integrated Vertex AI feature, registry, and training services. Used secure endpoints, load testing, monitored releases, and GitOps to support peak retail traffic.
- Leads the design and cross-functional delivery of AI/ML platforms; supported India AI hub growth through 300+ resume screens, 50+ technical interviews, and mentoring.

Lead ML Engineer - ML Engineering & Operations

Wipro | Aug 2023 - Jul 2025

- **AWS Account Digital Fingerprinting:** Built behavioral anomaly detection for AWS account security: trained unsupervised autoencoders per entity (account/user/service/machine) on NVIDIA DGX (8xA100) using NVIDIA Morpheus, learning normal activity fingerprints and surfacing deviations in real time. Used 20+ days of logs across multiple AWS accounts at billion-row scale per training run; integrated continuous retraining, MLflow model management, and Grafana.
- **Proofpoint Phishing Detection:** Engineered Morpheus training and inference pipelines with RAPIDS cuDF/CuPy, scoring 2,400 emails/sec, double the previous rate. Used security-team feedback and Microsoft Sentinel integration to detect phishing missed by initial filters and accelerate quarantine response.
- **Brand Infringement & Abuse-Inbox Analysis:** Built ML routing and OCR extraction of scammer identifiers across text, HTML, images, and attachments; routed evidence to security teams and mail partners for takedowns. Automated triage doubled the prior case volume and reduced manual review.
- Led ML/MLOps engineers, mentored team members, shared technical knowledge, and coordinated delivery with client and NVIDIA stakeholders.

Experience - continued

Senior Data Scientist - Data Science / Azure ML

Accenture | May 2021 - Sep 2023

- **Product Quality Prediction - BamaGruppen:** Built Azure ML warehouse-arrival quality prediction for the Norwegian distributor, with automated retraining, Logic Apps, Azure Pipelines, storage, and stakeholder alerts. Increased truck inspections from 130 to 160/day, defect detection by 17%, and customer approval by 20%, delivering \$2.5M annual savings.
- **Last-Mile Vehicle Routing:** Designed OR-Tools vehicle-routing models with capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.
- **Mellom Transport Optimization:** Built mixed-integer assignment models for daily terminal-to-distribution-center transport. Hosted APIs on Azure Function Apps; Logic Apps, RPA, CI/CD, and Slack notifications automated planning and dispatch, replacing manual Excel workflows.
- **Norsk Oppstrom - Simulation Framework:** Modelled product flows across vendors, warehouses, hubs, and customers, comparing cost, freshness, and carbon-footprint scenarios. Integrated simulation results with Power BI and identified the appropriate number and locations of regional hubs.
- **Hub Analytics POC - Network Impact Study:** Analyzed network flows and simulated strategic distribution centers to identify customers and vendors with the greatest benefit, assess warehouse-volume reduction, and present actionable hub-placement recommendations.
- Led cross-functional delivery, mentored data scientists, and presented findings and recommendations to Norwegian stakeholders.

Data Scientist

GD Research Centre Pvt Ltd | May 2020 - May 2021

- **GlobalData Forecasting:** Built domain-aware forecasts and scenario analysis across 15+ industry verticals, informing strategic market-intelligence reports and resource-allocation decisions.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

- **Semiconductor Defect Classification:** Built a distributed DenseNet training pipeline on about one million images with roughly three-hour training cycles. Deployed Docker/TensorFlow Serving inference at Amkor in South Korea, replacing manual visual inspection; trained the client team to operate the system.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

- **Opportunity Win/Lose Prediction:** Built logistic-regression scoring with SSIS data pipelines and Power BI/SSAS dashboards for data-driven sales prioritization.
- **Automated ID-Card Generation:** Delivered tested new-member insurance workflows, from impact analysis and design through implementation, support, and team knowledge sharing.

Open-Source Project

vLLM Model Chooser

Built an interactive GPU-to-model matchmaker with VRAM, quantization, and KV-cache fit filters. [Demo](#).

Certifications

- Google Cloud Certified: Generative AI Leader | Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Power BI Data Analyst Associate
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

Education

B.E. Electrical & Electronics

Andhra University, Visakhapatnam | 2012 - 2016 · 80%